

PAPER_966: Unified Triadic Solver

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Abstract

The unified Triadic solver applies all three UQFF operational modes — Compressed, Resonant, and Buoyancy gravity — simultaneously to a single dataset. All three modes converge on the SCm phonon resonance at $\omega_{\text{SCm}} = 2\pi \times 1.25$ THz.

1. Compressed Mode

$$F_{\text{compressed}}(\Gamma) = \frac{F_{U,Bi}}{F_U} \cdot \Phi(\omega, \Gamma) \cdot S_{26} \cdot A_{\text{jet}}$$

2. Resonant Mode

$$\Phi(\omega, \Gamma) = \Phi_0 \cdot \exp\left(-\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2}\right) \cdot S_{26}$$

3. Buoyancy Mode

$$E_{\text{net}}(t, \Gamma) = S_{26} \cdot \cos(\omega_{\text{SCm}} t) \cdot \exp(-\Gamma t)$$

4. Convergence

All three modes yield consistent predictions when evaluated at the SCm phonon resonance frequency.

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)